CMSC131 (03XX–04XX) Exam 1 Practice

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Introduction

Hey everyone!

This is a practice document for your upcoming **CMSC131 Exam 1 on October 7th**. The primary goal of this document is to help you prepare by reviewing important concepts and practicing the types of problems you may encounter.

However, completing every question in this document does **not** guarantee that you are fully prepared for the exam. Instead, use this as a preparation tool to help identify topics you understand well and areas where you may need some additional review.

As you work through the problems, try to understand `why` each solution works rather than simply memorizing answers. If you struggle with a question, use it as a sign that the corresponding topic may be worth reviewing again.

Good luck studying, and I hope this helps you prepare! :)



Short Answer Questions

Question 1. A java compiler takes _____ and returns _____

Question 2. What are the two types of comment in java and how do we write them?

Question 3. What is JVM? What makes it special?

Question 4. What is the main method? What is its purpose in a Java program?

Question 5. Which of the following are valid java identifiers in java?

`return totalValue float @name TotalValue _count HELLOWORLD String my.name hello!`

Question 6. What makes float and long unique? (Hint: think about how you declare them)

Question 7. Whats the difference between = and == in java

Question 8. What is String Concatenation in java?

Question 9. What happens when I do String str = 5 + 7;

Question 10. What class helps us take some input from the user?

Question 11. How do we initialize a variable to do so? (Follow up to Question 10.)

Question 12. What is the difference between next() and nextLine()?

Question 13. Susan tries to compare two strings using " == " but was stopped by their friend John. What could be the reason?

Question 14. Is it possible to write an else statement without an if statement?

Question 15. Whats the difference between writing multiple if statements back-to-back vs writing some else if statement? Does the program flow change?

Question 16. What is a scope?

Question 17. What is a local variable?

Question 18. Explain the difference between the logical operator (AND) and (OR).

Question 19. What is the naming convention for classes and packages in Java? Is it any different from the way we name variables?

Question 20. What is short circuiting?

Question 21. When do we use do-while loops?

Question 22. (TRUE/FALSE) All while loops can be written as for loops.

Question 23. How do we prevent infinite loops?

Question 24. Whats a nested loop?

Question 25. What type of method does not need to return a value?

Question 26. Explain the difference between public and private methods.

Question 27. What is operator precedence? How can parentheses change the order in which an expression is evaluated?

Question 28. What is the result of 7 / 2 in Java? What about 7.0 / 2? Explain why they are different.

Question 29. What is the difference between a parameter and an argument in a method?

Question 30. What does the final keyword do when declaring a variable?

Code Tracing and Debugging

For each of the following questions, carefully trace the code. Determine the exact output, identify any errors, or answer the question provided.

Question 1. What is the exact output of the following code?

```
int x = 5;
int y = 3;

System.out.println(x + y * 2);
System.out.println((x + y) * 2);
```

Question 2. What is the exact output of the following code?

```
String str = "CMSC";
int num = 131;

System.out.println(str + num);
System.out.println(num + 10 + str);
System.out.println(str + num + 10);
```

Question 3. Does the following code compile? If not, explain why and correct the error.

```
String name = "Java";
String firstLetter = name.charAt(0);

System.out.println(firstLetter);
```

Question 4. What is the exact output of the following code?

```
int x = 8;

if (x > 5) {
    System.out.print("A");
}

if (x > 10) {
    System.out.print("B");
} else {
    System.out.print("C");
}
```

Question 5. What is the exact output of the following code?

```
int x = 2;

while (x < 10) {
    System.out.print(x + " ");
    x += 2;
}
```

Question 6. What is the exact output of the following nested loop?

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print(j);
    }
    System.out.println();
}
```

Question 7. Does the following code compile? If not, explain why.

```
if (true) {
    int score = 100;
}

System.out.println(score);
```

Question 8. What is the exact output of the following code?

```
int x = 5;
int y = 2;

double result = x / y;

System.out.println(result);
```

Question 9. What is the exact output of the following code?

```
int x = 5;

System.out.println(x++);
System.out.println(x);
System.out.println(++x);
```

Question 10. Consider the following method:

```
public static int mystery(int x, int y) {
    if (x > y) {
        return x - y;
    }
    return x + y;
}
```

What is printed by the following code?

```
System.out.println(mystery(8, 3));
System.out.println(mystery(2, 7));
```

Question 11. What is the exact output of the following code?

```
String str = "Snoopy wishes you the best :>";

for (int i = 0; i < str.length(); i += 2) {
    System.out.print(str.charAt(i));
}
```

Question 12. Does the following code result in an error? If so, explain what causes the error.

```
String str = "Brooklyn Nets is the best team ever";
for (int i = 0; i <= str.length(); i++) {
    System.out.println(str.charAt(i));
}
```

Question 13. What is the exact output of the following code?

```
int x = 4;
int y = 0;

if (x > 2 || ++y > 0) {
    System.out.println("Hello");
}

System.out.println(y);
```

Explain why `y` has the value that it does.

Question 14. Find and correct the error in the following loop.

```
int x = 1;

while (x <= 5) {
    System.out.println(x);
}
```

Question 15. What is the exact output of the following code?

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= 2; j++) {
        System.out.print(i + "" + j + " ");
    }
}
```

Coding Questions

Question 1. Write a static method `reverse(String str)` that returns a string in reverse order.

Question 2. Write a static method `countLowerCaseVowels(String str)` that returns the number of lowercase vowels (`a, e, i, o, u`) in the string, excluding the first and last characters of the string. Uppercase vowels should not be counted. You may assume the string contains at least two characters.

Question 3. Write a static method `findCharacter(String str, char target)` that returns the index of the first occurrence of target. Return -1 if it does not occur.

Question 4. Write a static method `printRectangle(int rows, int columns)` that that prints a rectangle of (*) characters.

For example calling printRectangle(3,5); should print

```
*****
*****
*****
```

Question 5. Write a static method `winner(int score)` that takes two player names as input from the user using `Scanner`. The parameter `score` represents the number of points needed to win. Player 1 rolls first, followed by Player 2. Each roll generates a random integer from 1–6 which is added to that player's total. After each roll, check whether that player has reached or exceeded `score`. The game ends immediately when a player wins. Print the winner's name and final score.

Question 6. Write a static method `numberTriangle(int n)` that uses nested loops to print a number triangle with `n` rows. Each row should print the numbers from 1 up to the current row number. You may assume that `n > 0`.

For example, calling `numberTriangle(5)` should print:

```
1
12
123
1234
12345
```

A Final Note :)

I took CMSC131 with Pedram last semester, so I understand the situation a lot of you are in right now. My biggest piece of advice is: **please do not memorize the material**. Try to actually understand what each concept does, why we use it, and when we should use it. Knowing the syntax is important, but understanding the idea behind the syntax is much more useful.

For programming questions, take a moment to understand what the problem is actually asking before you start writing code. Trace through an example yourself and explain the problem to yourself. Think about what tools make sense: Do I need a `for` loop, a `while` loop, or a `do-while` loop? Can this be solved with an `if/else`? Do I need a nested loop? Breaking the problem into smaller pieces can make a question that initially looks difficult much easier.

Even though you will not be writing pseudocode on the exam, **practice thinking through your solution before coding**. If you get stuck on a question during the exam, do not panic or spend all your time staring at it. Move on and come back to it later. Sometimes coming back with a clear mind is all you need. You might even see something in another question that reminds you of an idea or gives you a small hint about how to approach the question you skipped.

When practicing, if your code is not working, try not to immediately use AI or look up the solution. **Trace your own code first.** Go through it iteration by iteration and keep track of how your variables are changing. Ask yourself what each line is doing and compare what you expected to happen with what is actually happening. Learning how to debug your own code will help you much more on the exam than simply seeing the correct solution (come to Office Hours if you feel stuck anywhere!!).

I also highly recommend looking through your projects again before the exam. Do not just look at whether you got them correct, rather look at the code you wrote and make sure you understand *why* it works. Your projects contain a lot of the concepts you have been practicing throughout the semester.

Exam logistics and past exams (I highly recommended past papers for practice):
https://www.cs.umd.edu/class/fall2026/cmsc131-03XX-04XX/exams/exam1/

Understand, don't memorize. Trace your code. Stay calm. You've got this! :)

Good luck on Exam 1!


YOU GOT
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